C23 Transition Metals – Knowledge Organiser

AQA A-level Chemistry • Set 1: Definitions and Ligands **Paper 1**

#	Question	Answer
1	Transition metal	Forms at least one stable ion with an incomplete d subshell
2	Four characteristic properties	Complexes, coloured ions, variable oxidation states, catalysis
3	Why Zn is not a transition metal	Zn^{2+} has a full d subshell
4	Ligand	Species that donates an electron pair to a metal ion
5	Complex	Central metal ion with coordinately bonded ligands
6	Coordination number	Number of coordinate bonds to the metal ion
7	Examples of monodentate ligands	H_2O, NH_3, Cl^-
8	Examples of bidentate ligands	Ethanedioate, ethane-1,2-diamine
9	Example of a multidentate ligand	$EDTA^{4-}$
10	Coordinate bonds formed by one $EDTA^{4-}$	6

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AQA A-level Chemistry • Set 2: Shapes and Isomerism **Paper 1**

#	Question	Answer
1	Shape of a complex with 6 small ligands	Octahedral, 90°
2	Shape of a complex with 4 Cl^- ligands	Tetrahedral, 109.5°
3	Why Cl^- gives coordination number 4	Larger ligand
4	Example of a square planar complex	Cisplatin
5	Example of a linear complex	$[Ag(NH_3)_2]^+$
6	Isomerism in square planar complexes	Cis–trans
7	Isomerism in octahedral complexes with 3 bidentate ligands	Optical
8	Chelate effect	Multidentate ligands replace monodentate; entropy increases
9	Metal ion in haem	Fe^{2+}
10	Why CO is toxic	Binds to haem more strongly than O_2

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AQA A-level Chemistry • Set 3: Colour **Paper 1**

#	Question	Answer
1	Cause of colour in complexes	d electrons absorb visible light and are excited
2	Energy–frequency equation	$\Delta E = h\nu = hc/\lambda$
3	Name of *h*	Planck constant
4	Three factors that change ΔE	Oxidation state, ligand, coordination number
5	Why Sc^{3+} and Zn^{2+} compounds are colourless	No d–d transitions possible (d^0 / d^{10})
6	Colour observed	Light not absorbed (transmitted)
7	Colorimetry measures	Absorbance
8	Filter colour chosen in colorimetry	Complementary colour
9	Calibration curve	Absorbance against known concentrations
10	Colour of $[CuCl_4]^{2-}$	Yellow

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AQA A-level Chemistry • Set 4: Variable Oxidation States **Paper 1**

#	Question	Answer
1	Colour of VO_2^+ (V +5)	Yellow
2	Colour of VO^{2+} (V +4)	Blue
3	Colour of V^{3+}	Green
4	Colour of V^{2+}	Violet
5	Reducing agent for vanadium(V) to vanadium(II)	Zinc in acid
6	Colour of $Cr_2O_7^{2-}$	Orange
7	Colour of CrO_4^{2-}	Yellow
8	Chromate–dichromate equilibrium	$2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$
9	Reagent to oxidise Cr^{3+} to CrO_4^{2-}	H_2O_2 in alkali
10	Product when $[Co(NH_3)_6]^{2+}$ is oxidised by air	$[Co(NH_3)_6]^{3+}$

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AQA A-level Chemistry • Set 5: Redox Titrations **Paper 1**

#	Question	Answer
1	MnO_4^- half-equation	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
2	$Cr_2O_7^{2-}$ half-equation	$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
3	Fe^{2+} half-equation	$Fe^{2+} \rightarrow Fe^{3+} + e^-$
4	$C_2O_4^{2-}$ half-equation	$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$
5	Mole ratio MnO_4^- : Fe^{2+}	1 : 5
6	Mole ratio MnO_4^- : $C_2O_4^{2-}$	2 : 5
7	End point of a $KMnO_4$ titration	First permanent pale pink
8	Acid used in $KMnO_4$ titrations	Dilute H_2SO_4
9	Why HCl is not used	Cl^- is oxidised by MnO_4^-
10	Why no indicator is needed with $KMnO_4$	Self-indicating

C23 Transition Metals – Knowledge Organiser

AQA A-level Chemistry • Set 6: Catalysts **Paper 1**

#	Question	Answer
1	Heterogeneous catalyst	Different phase from the reactants
2	Homogeneous catalyst	Same phase as the reactants
3	Why transition metals are good catalysts	Variable oxidation states
4	Purpose of a support medium	Increases surface area
5	Catalyst poisoning	Impurities block active sites
6	Contact process catalyst	V_2O_5
7	Contact process step 1	$SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$
8	Contact process step 2	$V_2O_4 + ½O_2 \rightarrow V_2O_5$
9	Haber process catalyst	Fe
10	Autocatalyst in the MnO_4^- / $C_2O_4^{2-}$ reaction	Mn^{2+}

C23 Transition Metals – Knowledge Organiser

AQA A-level Chemistry • Set 7: Homogeneous Catalysis **Paper 1**

#	Question	Answer
1	Catalyst for $S_2O_8^{2-}$ with I^-	Fe^{2+} (or Fe^{3+})
2	Why $S_2O_8^{2-}$ and I^- react slowly alone	Both negative, so repel
3	Catalysed step 1	$S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$
4	Catalysed step 2	$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$
5	Overall equation	$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$
6	Autocatalysis	A product catalyses the reaction
7	Mn^{2+} catalysis step 1	$4Mn^{2+} + MnO_4^- + 8H^+ \rightarrow 5Mn^{3+} + 4H_2O$
8	Mn^{2+} catalysis step 2	$2Mn^{3+} + C_2O_4^{2-} \rightarrow 2Mn^{2+} + 2CO_2$
9	Shape of rate curve for autocatalysis	Slow start, then speeds up
10	Why the catalyst is regenerated	Reduced then re-oxidised (or vice versa)